

Control Structures

In previous chapters, we have seen that a program is a set of statements separated by a semi colon. Also, these statements are executed sequentially (one after another) from top to bottom. However, at times, the program needs to be executed either a statement or a block at a time, depending on certain conditions. This is known as control structure. There are two concepts involved here - Branching and Looping.

4.1 Branching

The conditional execution of a statement or a group of statements is known as Branching. For these purpose C++ provides the following two methods.

4.1.1 The if Statement

The 'if' keyword is used to execute a statement or a group of statements with a specified condition.

The syntax for 'if' is as follows

```
if (condition)
    statement;
```

The statement will be executed when the condition is true, For example,

```
if (n > 10)
    cout << "Value of n = " << n;
```

In the above code, if the value of the variable 'n' is greater than 10, then the value of 'n' will be displayed on the screen.